

Hamiltonian-Inspired Energy Dissipation and Chunk-Wise Variational State Coupling for Efficient Multimodal State-Space Models

Madapati Naga Durga Abhinav Sai, Rajesh Duvvuru,
Avinash Paditapu, Konjet Premchand, T V R M Siva Shanmukh

School of Computer Science and Engineering, VIT-AP University,
Amaravati, 522237, Andhra Pradesh, India.

*Corresponding author(s). E-mail(s):

abhinav.24bca7148@vitapstudent.ac.in; rajesh.duvvuru@vitap.ac.in;

Abstract

State-Space Models (SSMs) offer sub-quadratic alternatives to self-attention, but deploying state-space recurrence in cross-modal vision-language metric learning raises two practical challenges: sensitivity to representation perturbations and potential misalignment between modality-specific boundary representations. We propose HEDO-HVSC, a multimodal state-space framework with two components: (i) the Hamiltonian-Inspired Energy Dissipation Operator (HEDO), a learned discrete coordinate-momentum transformation with positive damping that introduces a dissipative inductive bias, and (ii) Chunk-Wise Variational State Coupling (HVSC), which performs mask-weighted aggregation of chunk boundary states and regularizes modality-specific Gaussian latents through symmetric Kullback-Leibler (KL) divergence during training, while preserving independent, deterministic unimodal inference at test time. In a controlled 12-run factorial benchmark ($4 \text{ configurations} \times 3 \text{ seeds}$) on Flickr8k with frozen ViT-B/16 and RoBERTa-base backbones, the full model achieves a test Mean Recall of $48.28\% \pm 0.59\%$, comparable to a custom SSD-style baseline ($48.25\% \pm 0.47\%$; mean paired delta $+0.03\% \pm 0.20\%$). Corruption stress tests show a mixed robustness profile, with a $+0.50\%$ gain under severe Gaussian noise ($\sigma = 0.10$), and latency profiling over sequence lengths $L \in [16, 256]$ shows approximately linear empirical scaling.

Keywords: Image-text retrieval, State-space models, Hamiltonian-inspired dynamics, Variational representation learning, Vision-language alignment

1 Introduction

Cross-modal image-text retrieval is a foundational problem in multimodal machine learning, underpinning image search, visual question answering, and embodied agents [1–3]. The goal is to project visual patch sequences and textual token sequences into a joint metric space where semantic alignment corresponds to vector proximity [4, 5]. Transformer encoders built on self- and cross-attention are the standard foundation for vision-language models [6–8], but their cost grows quadratically, $\mathcal{O}(L^2)$, with sequence length [9], which becomes substantial for high-resolution images and long captions [10, 11].

Linear-time sequence models such as S4 [12], S5 [13], Mamba [14], and State-Space Duality (SSD) [15] mitigate this cost through discretized linear dynamical systems with $\mathcal{O}(L)$ complexity [16]. Applying state-space recurrence to multimodal metric learning, however, introduces two challenges. First, *representation perturbation sensitivity*: recurrent transitions propagate token variations, so localized noise, clutter, and photometric variation in visual patches can cause hidden-state drift [17, 18]. Second, *boundary representation misalignment*: separate unidirectional encoders may accumulate modality-specific drift, and unweighted pooling lets padding tokens influence final text representations [4, 19].

To investigate these challenges, we present HEDO-HVSC. Our contributions are:

1. **HEDO**, a learned discrete coordinate-momentum transformation ($\Delta t = 0.1$, $\beta = 0.05$, $\gamma = 0.1$) whose damping term limits momentum amplification prior to sequence recurrence.
2. **A state-continuous chunk-wise SSD-style recurrence** ($C = 16$, $d_{\text{state}} = 64$) in PyTorch that carries hidden states across chunk boundaries without resets.
3. **HVSC**, a mask-weighted boundary-state aggregation that parameterizes Gaussian latents regularized by symmetric KL divergence during training, while inference remains strictly unimodal and deterministic.
4. **A controlled multi-seed factorial benchmark** (12 runs) on Flickr8k [20] with frozen backbones, complemented by corruption testing and latency scaling analysis.

2 Related Work

Vision-language metric learning. VSE++ [4] and SCAN [5] established hard-negative ranking losses and stacked cross-attention for image-text matching. CLIP [1] and ALIGN [3] showed that dual encoders trained with symmetric InfoNCE on web-scale pairs yield robust representations, and BLIP [2] and BLIP-2 [10] added cross-attention decoders for fine-grained reasoning, at quadratic cost in sequence length.

Structured state space models. S4 [12] formalized continuous-time linear time-invariant systems with HiPPO initializations [16]; S5 [13] simplified it with parallel scans over diagonal state matrices. Mamba [14] made the recurrent parameters input-dependent, and SSD/Mamba-2 [15] established the equivalence between structured masked attention and recurrence. We use a custom PyTorch SSD-style recurrence inspired by this perspective rather than native Mamba-2/SSD CUDA kernels.

Hamiltonian and variational models. Hamiltonian [21] and Lagrangian [22] neural networks embed energy conservation into learning, and port-Hamiltonian systems [23] add positive semi-definite damping. VAEs [24] and the Variational Information Bottleneck [25] formalize latent representation learning. HEDO adapts discrete dissipative mechanics into a sequence operator, and HVSC uses Gaussian latents over recurrent boundary states for cross-modal alignment.

3 Problem Formulation and Background

Cross-modal metric learning. Let $\mathcal{D} = \{(\mathbf{I}_i, \mathcal{T}_i)\}_{i=1}^N$ be a dataset where each image $\mathbf{I}_i$ has M captions $\mathcal{T}_i = \{\mathbf{T}_{i,m}\}_{m=1}^M$. We learn encoders $f_\theta(\mathbf{I}), g_\phi(\mathbf{T}) \in \mathbb{S}^{d-1}$ such that semantic relevance correlates with cosine similarity $s(\mathbf{I}, \mathbf{T}) = f_\theta(\mathbf{I})^\top g_\phi(\mathbf{T})$.

State-space recurrence. A continuous SSM $\dot{\mathbf{h}}(t) = \mathbf{A}\mathbf{h}(t) + \mathbf{B}u(t)$, $y(t) = \mathbf{C}\mathbf{h}(t) + \mathbf{D}u(t)$ discretized with zero-order hold and step Δ gives

$$\mathbf{h}_t = \overline{\mathbf{A}}\mathbf{h}_{t-1} + \overline{\mathbf{B}}u_t, \quad y_t = \mathbf{C}\mathbf{h}_t + \mathbf{D}u_t, \quad (1)$$

with $\overline{\mathbf{A}} = \exp(\Delta\mathbf{A})$. SSD [15] shows that for scalar-times-identity $\mathbf{A}$ the recurrence admits structured block matrix computation. Our benchmark implementation uses an explicit sequential PyTorch recurrence.

Dissipative mechanics. For coordinates $\mathbf{q}$, momenta $\mathbf{p}$, Hamiltonian $\mathcal{H}(\mathbf{q}, \mathbf{p})$, and damping $\mathbf{D} \succ 0$, dissipative dynamics $\dot{\mathbf{q}} = \nabla_{\mathbf{p}}\mathcal{H}$, $\dot{\mathbf{p}} = -\nabla_{\mathbf{q}}\mathcal{H} - \mathbf{D}\nabla_{\mathbf{p}}\mathcal{H}$ satisfy $\dot{\mathcal{H}} = -\nabla_{\mathbf{p}}\mathcal{H}^\top \mathbf{D} \nabla_{\mathbf{p}}\mathcal{H} \leq 0$. This continuous-time property motivates our construction but does not imply monotonic energy decay for the discrete learned HEDO update; “energy dissipation” refers to the damping mechanism and its physical motivation.

4 Proposed HEDO-HVSC Architecture

Backbone features. As shown in Fig. 1, an image $\mathbf{I} \in \mathbb{R}^{3 \times 224 \times 224}$ and a caption $\mathbf{T}$ are encoded by frozen backbones and linearly projected:

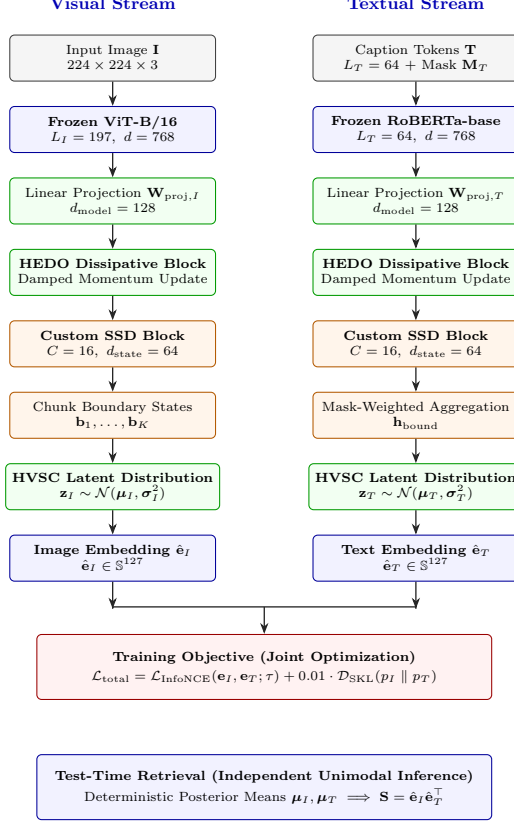
$$\mathbf{X}_I^{(0)} = \mathbf{W}_{\text{proj}, I} \text{ViT-B/16}(\mathbf{I}), \quad \mathbf{X}_T^{(0)} = \mathbf{W}_{\text{proj}, T} \text{RoBERTa}(\mathbf{T}), \quad (2)$$

with $\mathbf{X}_I^{(0)} \in \mathbb{R}^{B \times L_I \times d_{\text{model}}}$ ($L_I = 197$: 196 patches plus a class token), $\mathbf{X}_T^{(0)} \in \mathbb{R}^{B \times L_T \times d_{\text{model}}}$, and a text mask $\mathbf{M}_T \in \{0, 1\}^{B \times L_T}$. Both backbones remain frozen in all experiments.

4.1 Hamiltonian-Inspired Energy Dissipation Operator (HEDO)

Given input coordinates $\mathbf{q}_0 \in \mathbb{R}^{B \times L \times d_{\text{model}}}$, HEDO initializes momenta $\mathbf{p}_0 = \tanh(\mathbf{W}_p \mathbf{q}_0)$ and applies a damped coordinate-momentum update with $\Delta t = 0.1$, $\beta = 0.05$, and $\gamma = 0.1$:

$$\mathbf{p}_{t+1} = \mathbf{p}_t(1 - \beta\Delta t) - \Delta t \tanh(\mathbf{W}_q \mathbf{q}_t), \quad (3)$$



No Cross-Modal Attention | No Inter-Modality Hidden-State Exchange at Test Time

Fig. 1 HEDO-HVSC pipeline. Frozen ViT-B/16 ($L_I = 197$) and RoBERTa-base ($L_T = 64$) features are projected to $d_{\text{model}} = 128$, transformed by HEDO, and processed by chunk-wise SSD recurrence ($C = 16, d_{\text{state}} = 64$). During training, boundary states parameterize Gaussian latents regularized by $\mathcal{D}_{\text{SKL}}$ alongside multi-positive InfoNCE; at test time, deterministic posterior means are computed strictly unimodally.

$$\mathbf{q}_{t+1} = \mathbf{q}_t + \gamma \Delta t \mathbf{p}_{t+1}, \quad \text{HEDO}(\mathbf{q}) = \text{LayerNorm}(\mathbf{q}_{t+1}), \quad (4)$$

where $\mathbf{W}_p, \mathbf{W}_q \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$ are learnable. With the empirical energy $\mathcal{H}(\mathbf{q}, \mathbf{p}) = \frac{1}{2}(\|\mathbf{q}\|_2^2 + \|\mathbf{p}\|_2^2)$, update (3) satisfies

$$\|\mathbf{p}_{t+1}\|_2 \leq (1 - \beta \Delta t) \|\mathbf{p}_t\|_2 + \Delta t \|\tanh(\mathbf{W}_q \mathbf{q}_t)\|_2. \quad (5)$$

Since $\|\tanh(\cdot)\|_2 \leq \sqrt{d_{\text{model}}}$, the contraction factor $(1 - \beta \Delta t) = 0.995$ limits momentum amplification under the bounded force term. In pre-training diagnostics, HEDO reduced image representation energy from $\mathcal{H}_{\text{raw}} = 170.824$ to $\mathcal{H}_{\text{HEDO}} = 77.563$ while preserving cosine similarity 0.9952; however, unforced multi-step trajectories were not monotonically decreasing (images: $170.824 \rightarrow 174.387$ over six steps; text:

5.849 $\rightarrow$ 5.892). We therefore describe HEDO as a damping inductive bias, not a system with guaranteed monotonic energy decay.

4.2 State-Continuous Chunk-Wise SSD-Style Recurrence

The HEDO output $\mathbf{X}$ is projected to $d_{\text{state}} = 64$ by $\mathbf{B}_{\text{proj}}$, with learned decay $\mathbf{A}_{\text{decay}} = \exp(-\exp(\mathbf{A}_{\text{log}})) \in (0, 1)^{d_{\text{state}}}$. The sequence is split into $K = \lceil L/C \rceil$ chunks of size $C = 16$, and hidden states evolve as

$$\mathbf{h}_t = m_t (\mathbf{A}_{\text{decay}} \odot \mathbf{h}_{t-1} + \mathbf{B}_{\text{proj}} \mathbf{x}_t) + (1 - m_t) \mathbf{h}_{t-1}, \quad (6)$$

where $m_t \in \{0, 1\}$ marks valid tokens, so padded positions preserve the state. States carry across chunks ($\mathbf{h}_{\text{start},k} = \mathbf{h}_{\text{end},k-1}$, $\mathbf{h}_{\text{start},1} = \mathbf{0}$); the final state of each chunk is recorded as boundary state $\mathbf{b}_k$ with validity flag $m_k = \mathbb{I}(\sum_{t \in k} m_t > 0)$ (Fig. 2).

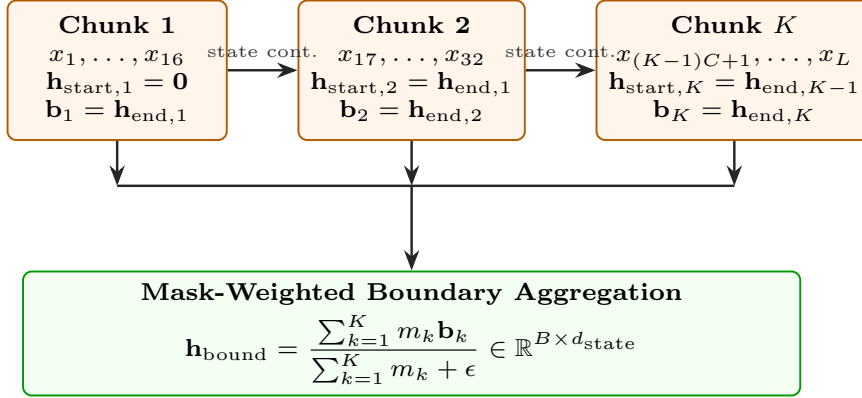


Fig. 2 State-continuous chunk-wise recurrence with boundary state extraction. Hidden states propagate across chunk boundaries without resets, and mask-weighted aggregation prevents padding bias.

4.3 Chunk-Wise Variational State Coupling (HVSC)

HVSC aggregates valid boundary states and parameterizes a diagonal Gaussian with $z_{\text{dim}} = 64$:

$$\mathbf{h}_{\text{bound}} = \frac{\sum_{k=1}^K m_k \mathbf{b}_k}{\sum_{k=1}^K m_k + \epsilon}, \quad (7)$$

$$\boldsymbol{\mu} = \mathbf{W}_{\boldsymbol{\mu}} \mathbf{h}_{\text{bound}}, \quad \log \boldsymbol{\sigma}^2 = \text{clamp}(\mathbf{W}_{\log \text{var}} \mathbf{h}_{\text{bound}}, -5, 2), \quad (8)$$

with $\epsilon = 10^{-6}$. During training, latents are sampled as $\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\epsilon} \odot \boldsymbol{\sigma}$ with $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. The sequence is mean-pooled (mask-weighted for text) into $\mathbf{h}_{\text{pool}}$ and fused as $\mathbf{e} = \text{LayerNorm}(\mathbf{h}_{\text{pool}} + \alpha \mathbf{W}_{\text{out}} \mathbf{z})$ with $\alpha = 0.1$. Cross-modal boundary alignment is regularized in FP32 by the symmetric KL divergence

$$\mathcal{D}_{\text{SKL}}(p_I \parallel p_T) = \frac{1}{2} [D_{\text{KL}}(p_I \parallel p_T) + D_{\text{KL}}(p_T \parallel p_I)], \quad (9)$$

computed analytically for diagonal Gaussians. At test time, sampling is omitted and the posterior means are used:

$$\hat{\mathbf{e}}_M = \text{Normalize}(\text{LayerNorm}(\mathbf{h}_{\text{pool},M} + \alpha \mathbf{W}_{\text{out}} \boldsymbol{\mu}_M)), \quad M \in \{I, T\}. \quad (10)$$

Visual and textual embeddings are thus computed independently, with no cross-modal attention or hidden-state exchange.

4.4 Training Objective

The loss is $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{InfoNCE}} + \lambda_{\text{KL}} \mathcal{D}_{\text{SKL}}$ with $\lambda_{\text{KL}} = 0.01$, where the multi-positive InfoNCE accounts for the 5 captions per image:

$$\mathcal{L}_{\text{InfoNCE}} = -\frac{1}{2B} \sum_{i=1}^B \left[\log \frac{\sum_{j \in \mathcal{P}(i)} e^{s_{i,j}/\tau}}{\sum_{k=1}^{5B} e^{s_{i,k}/\tau}} + \log \frac{e^{s_{i,i}/\tau}}{\sum_{k=1}^B e^{s_{k,i}/\tau}} \right], \quad (11)$$

where $s_{i,j} = \hat{\mathbf{e}}_{I,i}^\top \hat{\mathbf{e}}_{T,j}$, $\mathcal{P}(i)$ is the positive caption set of image i , and the learnable logit scale is capped at 100. Algorithm 1 summarizes one training step.

Algorithm 1 HEDO-HVSC forward pass and training step

Require: Images $\mathbf{I} \in \mathbb{R}^{B \times 3 \times 224 \times 224}$, tokens $\mathbf{T} \in \mathbb{Z}^{B \times 64}$, masks $\mathbf{M}_T$, positive index sets $\mathcal{P}$

Ensure: Total loss $\mathcal{L}_{\text{total}}$, test embeddings $\hat{\mathbf{e}}_I, \hat{\mathbf{e}}_T$

- 1: Extract frozen features $\mathbf{X}_I^{(0)}, \mathbf{X}_T^{(0)}$ via (2)
 - 2: Apply the HEDO transformation, if enabled, via (3)–(4)
 - 3: Run state-continuous chunk-wise SSD recurrence; collect boundary states $\mathbf{b}_k$ and flags m_k
 - 4: Aggregate boundary states and parameterize latents via (7)–(8)
 - 5: Compute $\mathcal{D}_{\text{SKL}}(p_I \parallel p_T)$ in FP32 via (9)
 - 6: Sample $\mathbf{z}$ and form coupled embeddings $\mathbf{e}_I, \mathbf{e}_T$
 - 7: Compute $\mathcal{L}_{\text{InfoNCE}}$ via (11)
 - 8: **return** $\mathcal{L}_{\text{total}} \leftarrow \mathcal{L}_{\text{InfoNCE}} + 0.01 \cdot \mathcal{D}_{\text{SKL}}$
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5 Experimental Setup

Dataset. We use Flickr8k [20]: 8,000 images with 5 captions each, split into 6,000 / 1,000 / 1,000 images (30,000 / 5,000 / 5,000 captions) for training, checkpoint selection, and held-out testing. Disjointness of the splits was verified via image identifier hashing.

Backbones and implementation. The visual backbone is the `timm` checkpoint `vit_base_patch16_224` (86.6 M parameters, ImageNet normalization) and the text backbone is `roberta-base` (124.6 M parameters, $L_T = 64$); both are frozen, and only projections, HEDO, SSD, and HVSC modules (0.330–0.422 M parameters) are trained. We use Python 3.12, PyTorch 2.10.0 (CUDA 12.8), `timm` 1.0.26, and `transformers`

Table 1 Hyperparameter specification.

Hyperparameter	Value	Hyperparameter	Value
Image resolution	224×224	Optimizer	AdamW (0.9, 0.999; wd 10^{-4})
$d_{\text{model}} / d_{\text{state}}$	128 / 64	Batch size	64 images (320 captions)
Chunk size C	16	Epochs	8
HEDO $\Delta t / \beta / \gamma$	0.1 / 0.05 / 0.1	LR schedule	Cosine, $2 \times 10^{-4} \rightarrow 10^{-6}$
HVSC latent dim	64	Gradient clipping	Norm 1.0
KL weight λ_{KL}	0.01 (FP32)	Seeds	42, 43, 44

Table 2 Retrieval results on the Flickr8k test split (1,000 images, 5,000 captions): mean (top) and standard deviation (bottom) over 3 seeds.

Model	Image-to-Text (%)			Text-to-Image (%)			MR (%)
	R@1	R@5	R@10	R@1	R@5	R@10	
SSD Baseline	26.53 ± 0.33	57.30 ± 0.80	70.10 ± 1.22	20.45 ± 0.56	49.89 ± 0.19	65.20 ± 0.43	48.25 ± 0.47
w/o HEDO	26.70 ± 0.82	57.07 ± 0.85	70.03 ± 1.11	20.29 ± 0.36	49.95 ± 0.21	65.27 ± 0.32	48.22 ± 0.43
w/o HVSC	27.20 ± 0.99	56.97 ± 0.19	70.30 ± 0.45	20.66 ± 0.53	50.33 ± 0.22	65.10 ± 0.24	48.43 ± 0.29
Full HEDO-HVSC	26.73 ± 1.87	56.83 ± 0.47	70.47 ± 0.80	20.71 ± 0.18	49.97 ± 0.40	64.96 ± 0.34	48.28 ± 0.59

5.0.0 on an NVIDIA Tesla T4 (16 GB), with AMP FP16 and gradient scaling. Checkpoints are selected by peak validation Mean Recall (patience 3 epochs, $\Delta_{\min} = 0.05\%$), and the test split is evaluated once. Table 1 lists the hyperparameters.

Protocol and metrics. Four configurations are evaluated with three seeds each (12 runs): (i) Custom SSD-Style Baseline, (ii) w/o HEDO, (iii) w/o HVSC, and (iv) Full HEDO-HVSC. Image-to-text (I2T) R@K ranks all 5,000 test captions per image and succeeds if any ground-truth caption is in the top K ; text-to-image (T2I) R@K ranks all 1,000 test images per caption. Mean Recall (MR) averages the six recalls. For robustness, a 100-image (500-caption) test subset was fixed a priori with seed 42 and evaluated under identical corruptions for all models; Gaussian noise was added pixel-wise with a fixed seed before clipping to $[0, 1]$.

6 Results

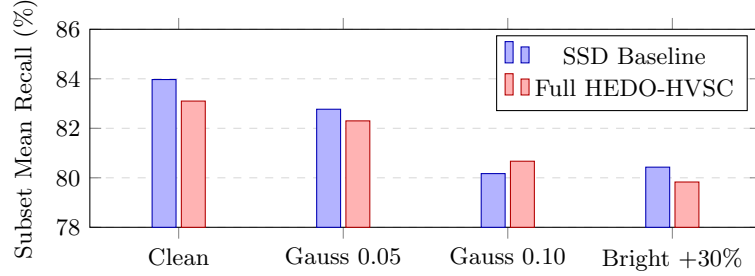
Main benchmark. Table 2 reports the aggregated test results. Full HEDO-HVSC achieves a Mean Recall of $48.28\% \pm 0.59\%$ (I2T R@1 26.73%, T2I R@1 20.71%), comparable to the baseline ($48.25\% \pm 0.47\%$); the w/o HVSC ablation achieves the highest Mean Recall ($48.43\% \pm 0.29\%$). Matched-seed deltas of the full model over the baseline are -0.14% (seed 42), -0.08% (seed 43), and $+0.32\%$ (seed 44), for a mean paired delta of $+0.03\% \pm 0.20\%$. The 2×2 factorial interaction, $\Delta_{\text{int}} = \text{MR}_{11} - \text{MR}_{10} - \text{MR}_{01} + \text{MR}_{00} = 48.28 - 48.43 - 48.22 + 48.25 = -0.12\%$, is small, indicating limited interaction between HEDO and HVSC on clean data.

Table 3 Corruption stress test (Mean Recall, 100-image subset).

Condition	Base	Full	Δ
Clean	83.97	83.10	-0.87
Gaussian $\sigma = 0.05$	82.77	82.30	-0.47
Gaussian $\sigma = 0.10$	80.17	80.67	+0.50
Brightness +30%	80.43	79.83	-0.60

Table 4 Inference latency of the SSD block ($B = 16$).

L	Latency (ms)	Relative
16	2.894 ± 0.021	$1.00\times$
32	5.436 ± 0.034	$1.88\times$
64	10.315 ± 0.052	$3.56\times$
128	19.713 ± 0.089	$6.81\times$
256	38.602 ± 0.141	$13.34\times$

**Fig. 3** Robustness under pixel-space perturbations on the 100-image test subset: the full model is better only under severe Gaussian noise ($\sigma = 0.10$).

Robustness. To examine whether dissipation attenuates perturbations, the baseline and full model were evaluated on the fixed 100-image subset under pixel-space corruptions (Table 3, Fig. 3). The profile is mixed: the full model is +0.50% better under severe Gaussian noise ($\sigma = 0.10$) but lower on clean images (-0.87%), mild noise (-0.47%), and brightness increase (-0.60%). This subset experiment is a secondary diagnostic rather than a comprehensive robustness evaluation.

Scaling and footprint. Latency was measured for $L \in \{16, \dots, 256\}$ ($d_{\text{model}} = 128$, $d_{\text{state}} = 64$) after 10 warm-up iterations, averaged over 50 timed iterations with explicit CUDA synchronization (Table 4). A linear fit, $\text{Latency}(L) = 0.149L + 0.582$ ms ($R^2 = 0.9998$), characterizes the measured range; the $\mathcal{O}(L)$ complexity follows from the sequential recurrence itself. The full model has 0.422 M trainable parameters (projections 46.7%, SSD blocks 23.3%, HVSC 22.4%, HEDO 7.8%, normalization 0.2%), under 0.21% of the 209.854 M frozen backbone parameters, with 2686.25 MB peak training VRAM (baseline: 0.330 M, 1878.22 MB; w/o HEDO: 0.356 M, 1878.81 MB; w/o HVSC: 0.397 M, 2685.66 MB).

7 Discussion

The full model matches the baseline on clean retrieval (paired delta $+0.03\% \pm 0.20\%$). Because only three seeds were evaluated per configuration, variability is reported descriptively rather than through inferential tests. The w/o HVSC configuration attains the highest Mean Recall, so the experiments do not show that HVSC improves clean-data accuracy; its role is best interpreted as a training-time cross-modal

boundary-distribution regularizer, with mask weighting preventing padding from contaminating boundary states. HEDO yields a small advantage under severe Gaussian noise but not across all corruptions.

What the results do not establish. The results do not establish state-of-the-art accuracy over fine-tuned, unconstrained dual encoders; HEDO does not provide a monotonic Lyapunov stability guarantee for discrete learned dynamics; robustness is not universally superior; the PyTorch recurrence is not compared against native CUDA Mamba-2 kernels; and scaling is measured only for $L \in [16, 256]$.

Limitations and future work. Evaluation is limited to Flickr8k; extending it to Flickr30k [26] and MS-COCO [27] is future work. Backbones were frozen, and joint fine-tuning may increase capacity. Compiling the recurrence into native CUDA kernels should further accelerate inference.

8 Conclusion

We introduced HEDO-HVSC, a multimodal state-space framework combining a Hamiltonian-inspired discrete dissipative operator with mask-weighted chunk-wise variational state coupling. In a 12-run controlled factorial benchmark on Flickr8k, HEDO-HVSC achieves retrieval performance comparable to a custom SSD-style baseline ($48.28\% \pm 0.59\%$ Mean Recall). Measured latency scales approximately linearly over the tested sequence lengths, consistent with the linear recurrent structure, and robustness evaluation shows a small advantage under severe Gaussian noise with mixed results under other perturbations.

References

- [1] Radford, A., Kim, J.W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., et al.: Learning transferable visual models from natural language supervision. Proceedings of the 38th International Conference on Machine Learning (ICML), 8748–8763 (2021)
- [2] Li, J., Li, D., Xiong, C., Hoi, S.: BLIP: Bootstrapping language-image pre-training for unified vision-language understanding and generation. Proceedings of the 39th International Conference on Machine Learning (ICML), 12888–12900 (2022)
- [3] Jia, C., Yang, Y., Xia, Y., Chen, Y.-T., Parekh, Z., Pham, H., Le, Q., Ku, S.-Y., Wang, J., Duerig, T.: Scaling up visual and vision-language representation learning with noisy text supervision. Proceedings of the 38th International Conference on Machine Learning (ICML), 4904–4916 (2021)
- [4] Faghri, F., Fleet, D.J., Kiros, J.R., Fidler, S.: VSE++: Improving visual-semantic embeddings with hard negatives. British Machine Vision Conference (BMVC) (2018)
- [5] Lee, K.-H., Chen, X., Hua, G., Houdusse, H., Vassilvitskii, S.: Stacked cross attention for image-text matching. Proceedings of the European Conference on

Computer Vision (ECCV), 201–216 (2018)

- [6] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, L., Polosukhin, I.: Attention is all you need. *Advances in Neural Information Processing Systems (NeurIPS)* **30** (2017)
- [7] Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., et al.: An image is worth 16x16 words: Transformers for image recognition at scale. *International Conference on Learning Representations (ICLR)* (2021)
- [8] Liu, Y., Ott, M., Goyal, N., Du, J., Joshi, M., Chen, D., Levy, O., Lewis, M., Zettlemoyer, L., Stoyanov, V.: RoBERTa: A robustly optimized BERT pretraining approach. *arXiv preprint arXiv:1907.11692* (2019)
- [9] Dao, T., Fu, D.Y., Ermon, S., Rudra, A., Ré, C.: FlashAttention: Fast and memory-efficient exact attention with IO-awareness. *Advances in Neural Information Processing Systems (NeurIPS)* **35**, 16344–16359 (2022)
- [10] Li, J., Li, D., Savarese, S., Hoi, S.: BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. *Proceedings of the 40th International Conference on Machine Learning (ICML)*, 19730–19742 (2023)
- [11] Wang, W., Bao, H., Dong, L., Bjorck, J., Peng, Z., Liu, Q., Aggarwal, K., Mohammed, O.K., Singhal, S., Som, S., et al.: Image as a foreign language: BEiT-3 pretraining for vision-language tasks. *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 19175–19186 (2023)
- [12] Gu, A., Goel, K., Ré, C.: Efficiently modeling long sequences with structured state spaces. *International Conference on Learning Representations (ICLR)* (2022)
- [13] Smith, J.T.H., Warrington, A., Linderman, S.W.: Simplified state space layers for sequence modeling. *International Conference on Learning Representations (ICLR)* (2023)
- [14] Gu, A., Dao, T.: Mamba: Linear-time sequence modeling with selective state spaces. *arXiv preprint arXiv:2312.00752* (2023)
- [15] Dao, T., Gu, A.: Transformers are SSMS: Generalized models and efficient algorithms through structured state space duality. *Proceedings of the 41st International Conference on Machine Learning (ICML)*, PMLR **235**, 10041–10071 (2024)
- [16] Gu, A., Dao, T., Dao, S., Ermon, S., Ré, C.: HiPPO: Recurrent memory with optimal polynomial projections. *Advances in Neural Information Processing Systems (NeurIPS)* **33**, 1474–1487 (2020)

- [17] He, K., Zhang, X., Ren, S., Sun, J.: Deep residual learning for image recognition. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 770–778 (2016)
- [18] Hendrycks, D., Dietterich, T.: Benchmarking neural network robustness to common corruptions and perturbations. International Conference on Learning Representations (ICLR) (2019)
- [19] Devlin, J., Chang, M.-W., Lee, K., Toutanova, K.: BERT: Pre-training of deep bidirectional transformers for language understanding. North American Chapter of the Association for Computational Linguistics (NAACL), 4171–4186 (2019)
- [20] Hodosh, M., Young, P., Hockenmaier, J.: Framing image description as a ranking task: Data, models and evaluation metrics. Journal of Artificial Intelligence Research **47**, 853–899 (2013) <https://doi.org/10.1613/jair.3994>
- [21] Greydanus, S., Dzamba, M., Yosinski, J.: Hamiltonian neural networks. Advances in Neural Information Processing Systems (NeurIPS) **32** (2019)
- [22] Cranmer, M., Greydanus, S., Hoyer, S., Battaglia, P., Spergel, D., Ho, S.: Lagrangian neural networks. ICLR Workshop on Deep Differential Equations (2020)
- [23] Massaroli, S., Poli, M., Califano, F., Faragasso, A., Park, J., Yamashita, A., Asama, H.: Port-hamiltonian approach to neural network training. In: Proceedings of the 2019 IEEE 58th Conference on Decision and Control (CDC), pp. 6799–6806 (2019). <https://doi.org/10.1109/CDC40024.2019.9030017>
- [24] Kingma, D.P., Welling, M.: Auto-encoding variational Bayes. International Conference on Learning Representations (ICLR) (2014)
- [25] Alemi, A.A., Fischer, I., Dillon, J.V., Murphy, K.: Deep variational information bottleneck. International Conference on Learning Representations (ICLR) (2017)
- [26] Young, P., Lai, A., Hodosh, M., Hockenmaier, J.: From image descriptions to visual denotations: New similarity metrics for semantic inference over event descriptions. Transactions of the Association for Computational Linguistics (TACL) **2**, 67–78 (2014) https://doi.org/10.1162/TACL_A.00166
- [27] Lin, T.-Y., Maire, M., Belongie, S., Hays, J., Perona, P., Ramanan, D., Dollár, P., Zitnick, C.L.: Microsoft COCO: Common objects in context. Proceedings of the European Conference on Computer Vision (ECCV), 740–755 (2014)